SMART CANE
OBSTACLE DETECTION

Embedded Systems Project

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PROBLEM

White canes used by visually impaired members of society detect ground-level obstacles, but nothing above.

Consequent hazards of indoor navigation:
Objects such as :

- tables, shelves, open cabinets and other obstacles at body height remain completely undetected.

- Only one direction can be explored at a time. (the direction the cane is directed towards)

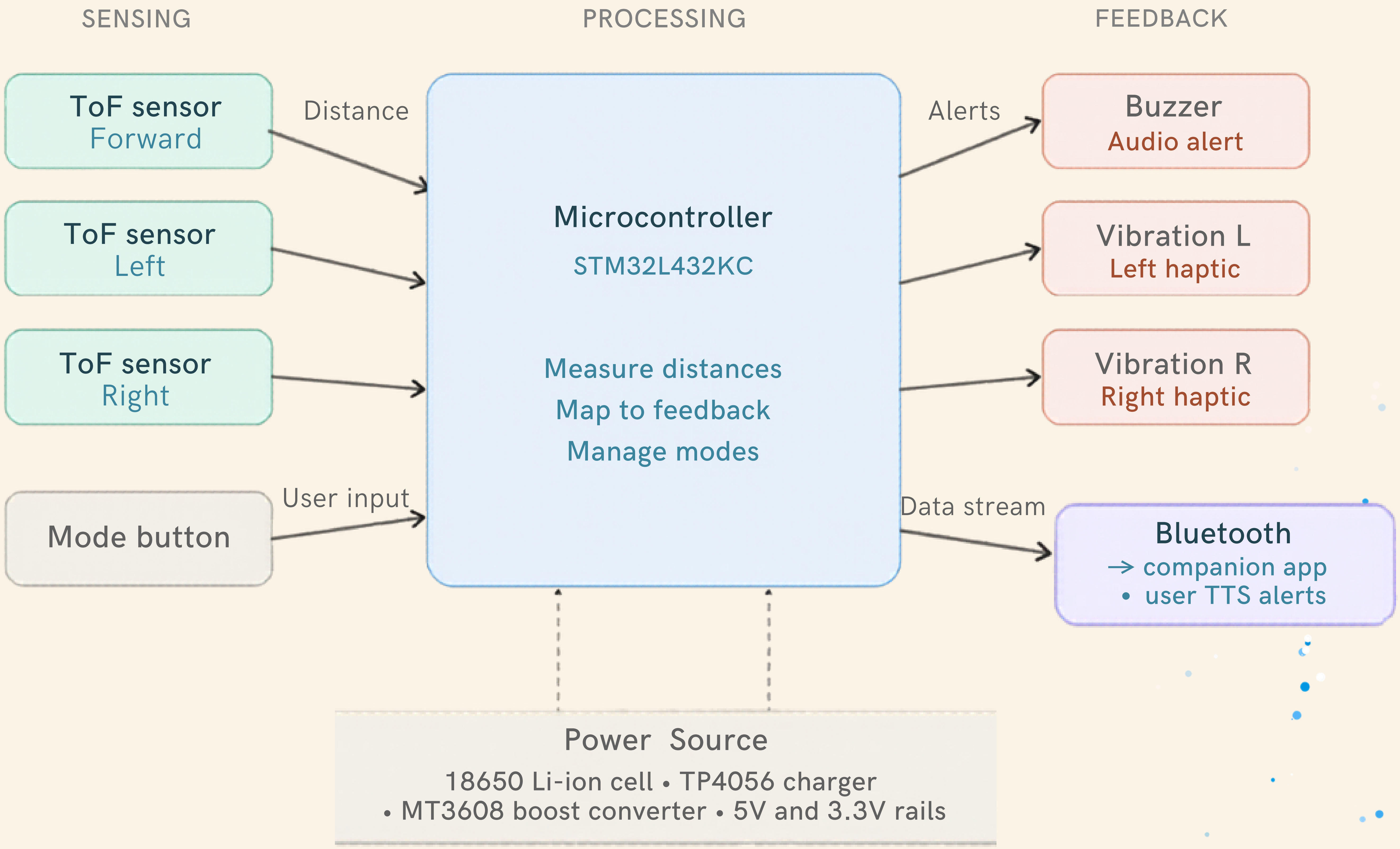
SOLUTION

We aim to create a clip-on attachment for the upper grip of the standard white cane, extending its capability beyond the ground to detect obstacles at body height in real time. Four feedback modes (buzzer-only, vibration-only, combined, neither/silent) let the user adapt to any environment.

HOW IT WORKS

- 3 VL53L1X laser Time-of-Flight sensors continuously scan for obstacles at body height (forward, left, and right)
- Audio feedback via distinct beep patterns per direction, with beep speed indicating distance + companion app converting obstacle data to spoken alerts via Text-to-Speech through the user's own earphones or headset
- Bluetooth streams obstacle data to companion app for TTS alerts
- Fully integrated 18650 rechargeable battery providing 8–12 hours of continuous use

BLOCK DIAGRAM



Main COMPONENTS

Component	Quantity
NUCLEO-L432KC	1
VL53L1X ToF Sensor	3
HM-10 Bluetooth Module	1
Buzzer	1
Coin Vibration Motor	2
Push Button	1
18650 Li-ion Cell	1
TP4056 Charging Module	1
MT3608 Boost Converter	1
Rocker Switch	1

FUNCTIONAL REQUIREMENTS

1. **Detect obstacles within 2m (forward, left, and right)**
2. **Deliver distinct beep patterns per direction, speed increasing with proximity alongside TTS alerts through companion app when phone is paired**
3. **Support four feedback modes switchable via push button**
4. **Transmit live distance data to a paired phone via Bluetooth**
5. **Poll all three sensors without interference**
6. **Response time under 500ms at walking pace**

CHALLENGES

1. **VL53L1X is relatively expensive**

2. **Sensor range decreases in direct sunlight (still within 2 metres) and is less accurate with reflective materials**

3. **Distinguishable feedback such that beep patterns must be distinct enough for a user to reliably tell directions apart**

4. **Power constraints: device must run long enough on a portable power source to be practically usable**

PLAN

Week 1

Setup & Individual Subsystems
Single sensor test, HM-10 link, power path validation

Week 2

Sensor Integration
Three sensors on I2C bus, buzzer + vibration PWM, start companion app

Week 3

Full System Integration
Full system integration, mode switching, TTS alerts connected

Week 4

Testing & Demo
Tune thresholds, fix bugs, prepare demo and finalize wiki

THANK YOU

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